



AI Semantic
Engine



Security &
Isolation

Librika (e-Pathshala)

The Next-Gen Smart Library Blueprint



Dynamic
Routing



Gamified
Logic

Unified System Architecture & Security

Built for speed and concurrency, Librika routes traffic dynamically while protecting persistent records.

Tech Stack

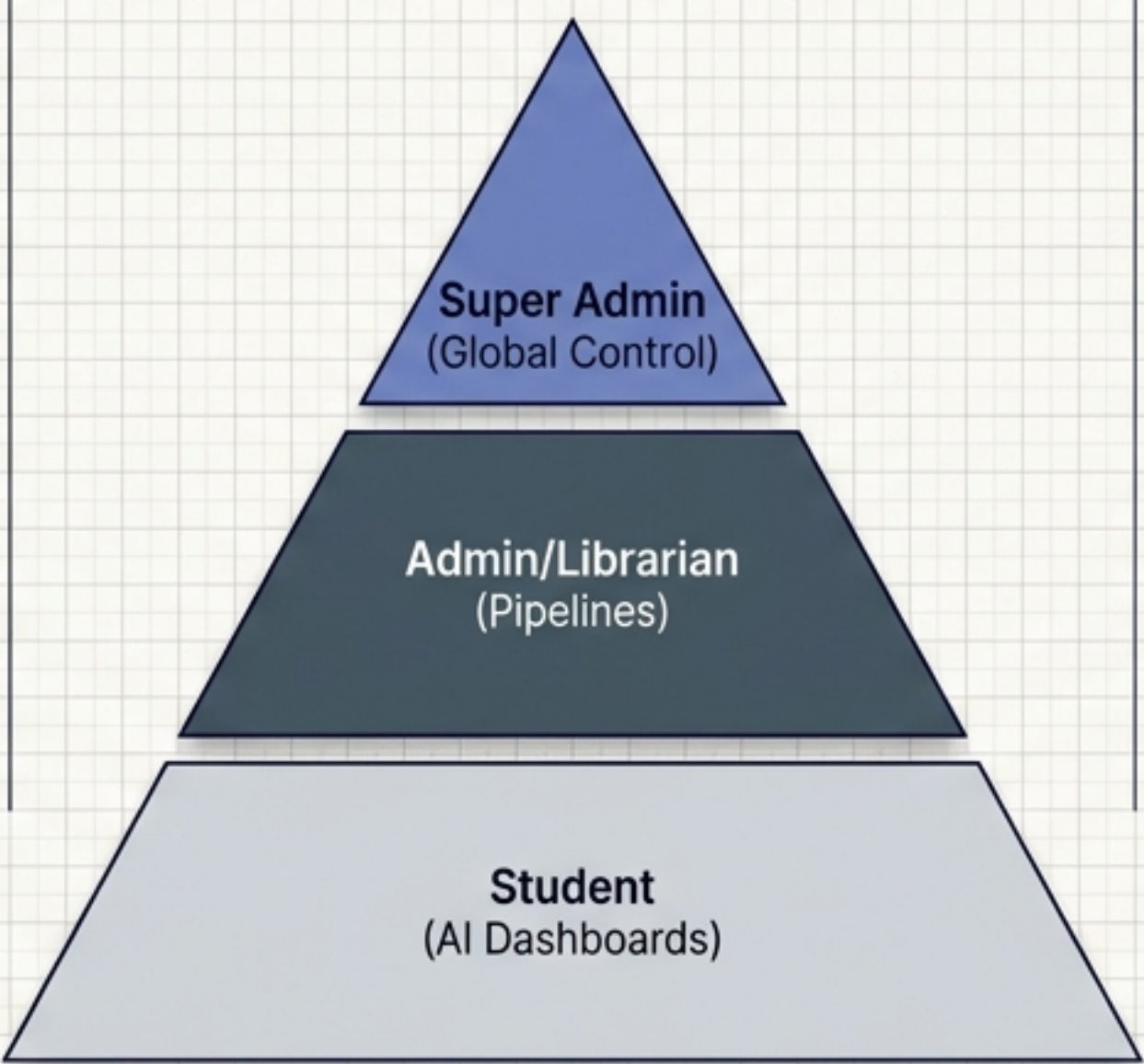
 Python Flask Backend

 SQLite + WAL Mode
(High Concurrency)

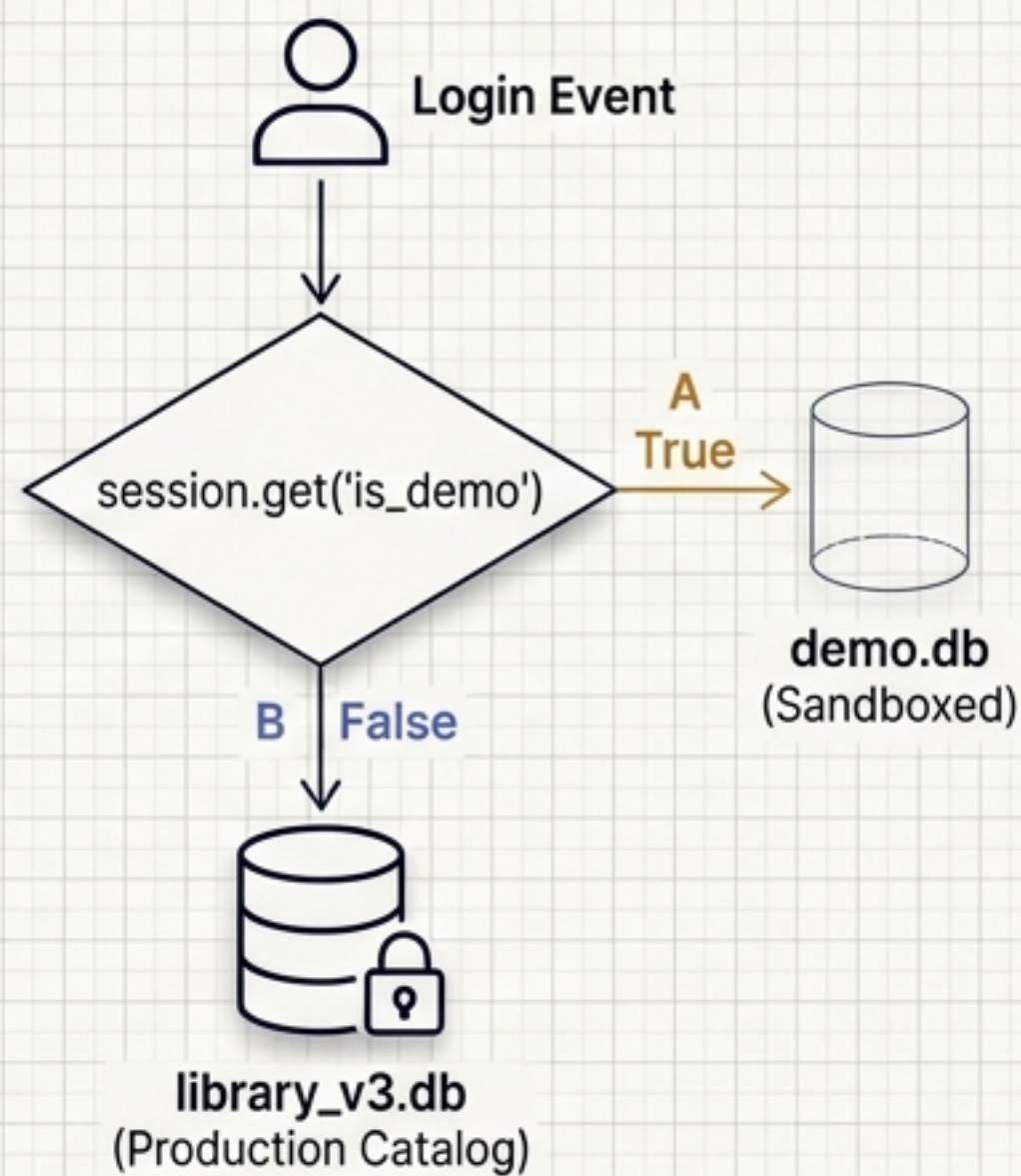
 Jinja2 Rendering

 Vanilla CSS
(Outfit & Plus Jakarta Sans)

Role-Based Access

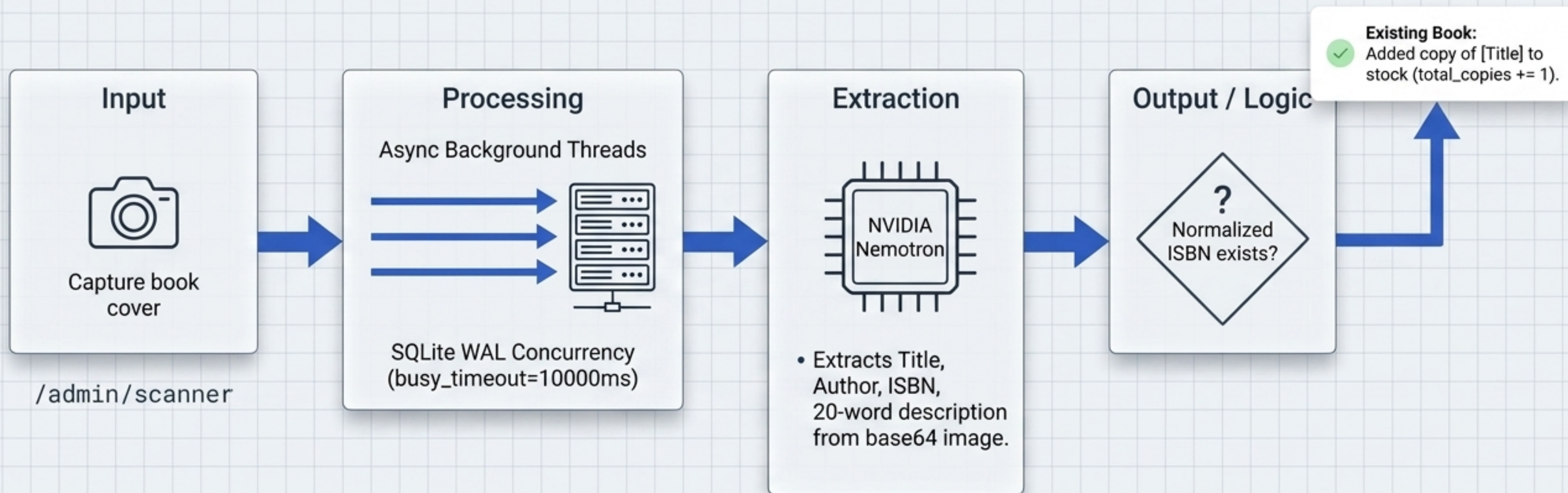


Dynamic Database Isolation



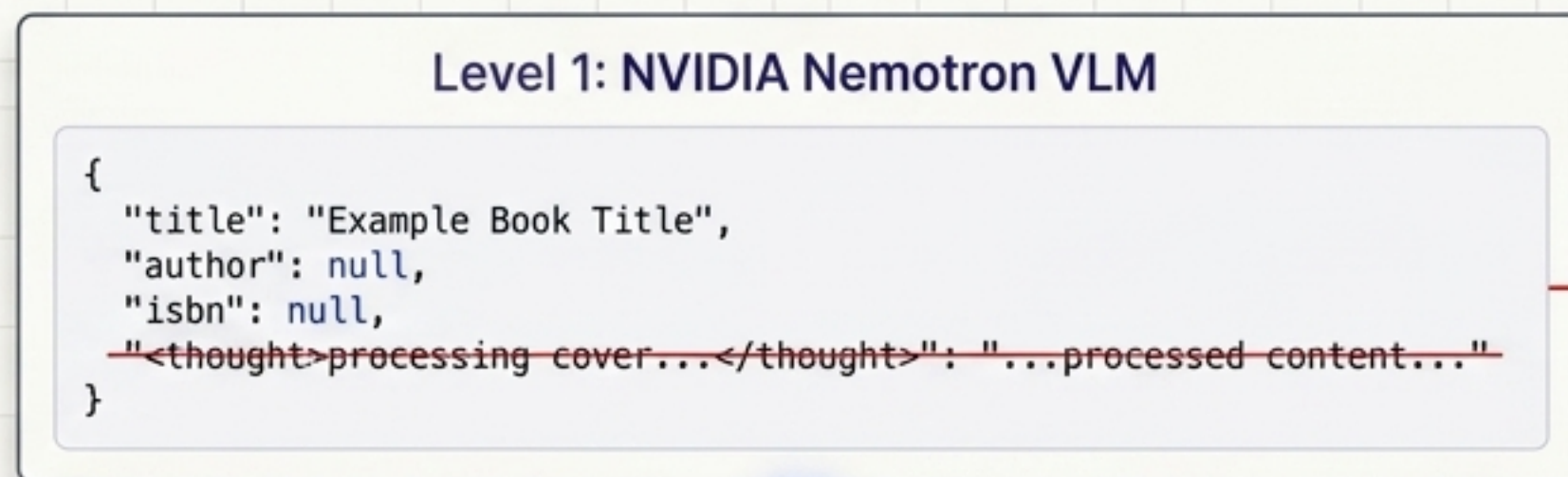
High-Speed AI Book Scanner

Performance: 10 Books Processed in 5 Seconds



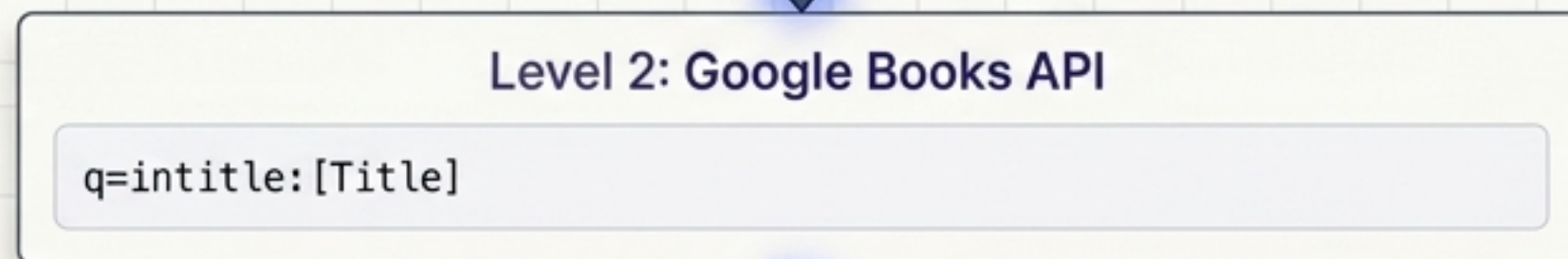
The Double Fallback Accuracy Engine

Rescuing incomplete metadata through multi-tiered fallback querying.



System automatically
strips thought blocks

If missing ISBN/Author...



If regional textbook / empty return...



Subscription Matrix & Interactive Billing

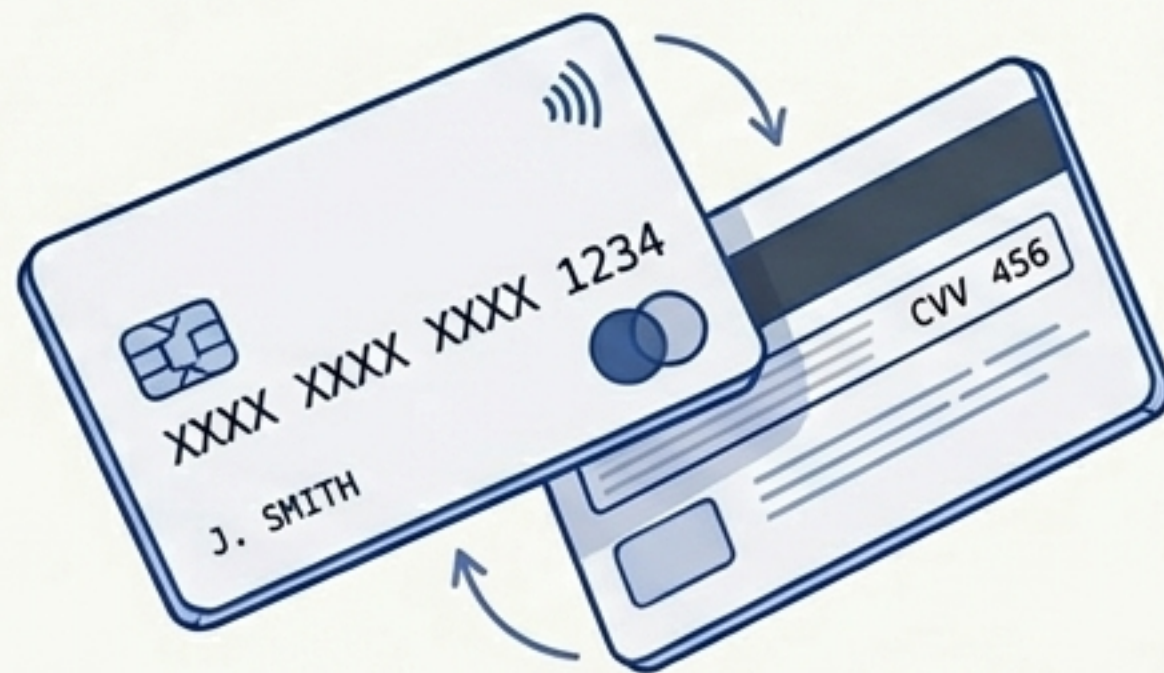
The Logic Matrix

Tier	Students	Catalog Books	AI Features
FREE	50	500	Disabled
BASIC	500	5,000	Enabled
PROFESSIONAL	Unlimited	Unlimited	Enabled

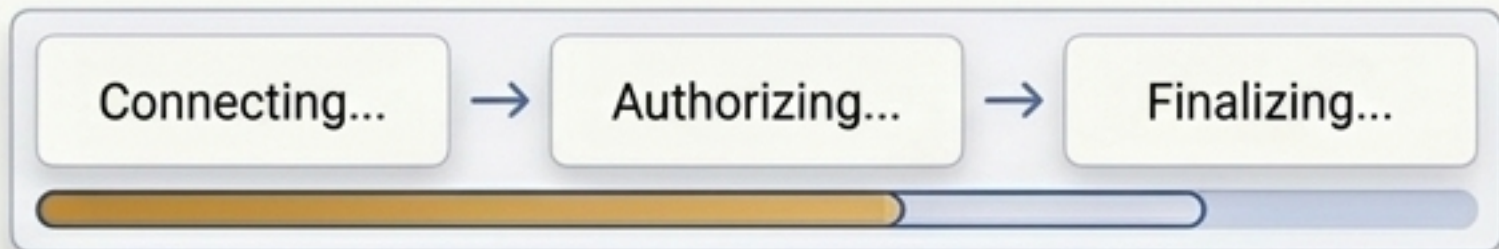


Limits dynamically queried on every add-user/add-copy action.

3D Checkout Simulator



Authorization Loader

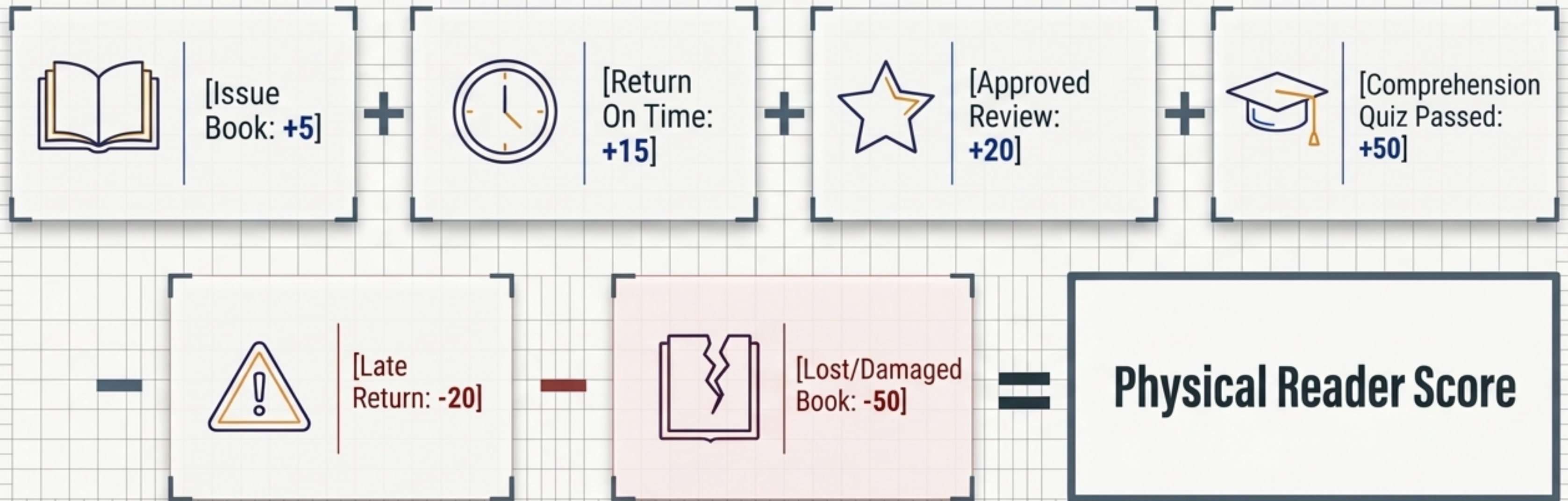


Cancel

Instant downgrade to FREE tier limits.

The Gamified Reader Score

Calculating engagement, comprehension, and asset care.



Dynamic Leaderboards & Achievements

/leaderboard

Filter Bar

Weekly

Monthly

Yearly

All-Time

Class Scope

Section

Physical Library
Checkouts

1st: Student A - 150

2nd: Student B - 120

3rd: Student C - 100

Digital Library
e-Library Uploads

1st: Student D - 85

2nd: Student E - 70

3rd: Student F - 60

Overall Leaderboard
Aggregated

1st: Student G - 235

2nd: Student H - 210

3rd: Student I - 190

Student Profile Hub

Comprehension
Champion

Reading Streak
Master

Digital
Contributor

**Daily Reading
Streak: 14 Days**

AI Comprehension Quiz System

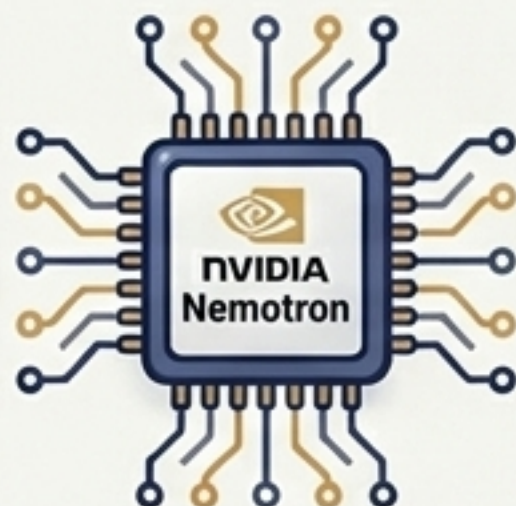
Generating on-demand assessments to verify learning and prevent gamification farming.

Step 1: Return



Student returns a completed physical book.

Step 2: Trigger



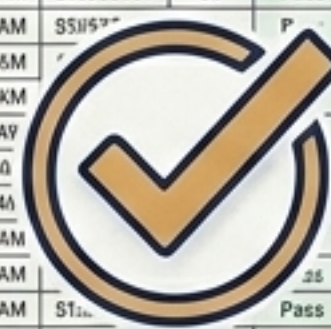
System triggers LLM completions via backend API.

Step 3: Generate



AI reads catalog metadata and instantly generates a 5-question multiple-choice quiz testing core concepts.

Step 4: Evaluate



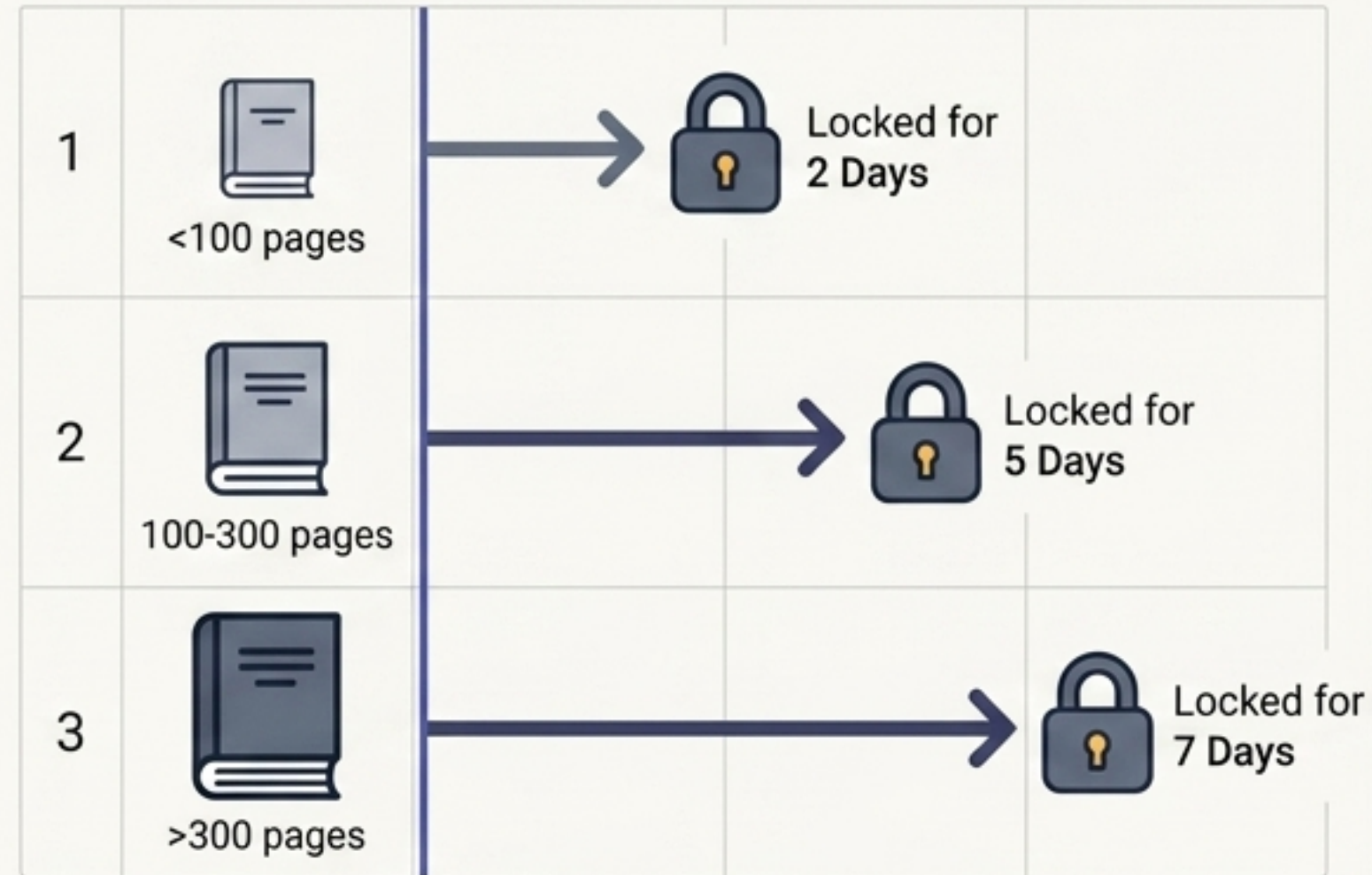
Time	Student ID	Quiz	Results	Score
9:47:09 AM	S50388951	61	Pass	236
9:42:28 AM	S511577		P	256
9:21:29 AM				235
9:22:38 AM				210
9:33:36 AM				190
5:22:20 AM				150
9:35:58 AM				160
9:33:36 AM				150
9:22:36 AM			Pass	190
9:22:36 AM	S1...		Pass	150
9:25:35 AM	801308491	03	Fail	150
9:23:18 AM	910950401	106	Fail	150

Student submits. System evaluates, logs Pass/Fail status, correct counts, and timestamp. Award +50 points on pass.

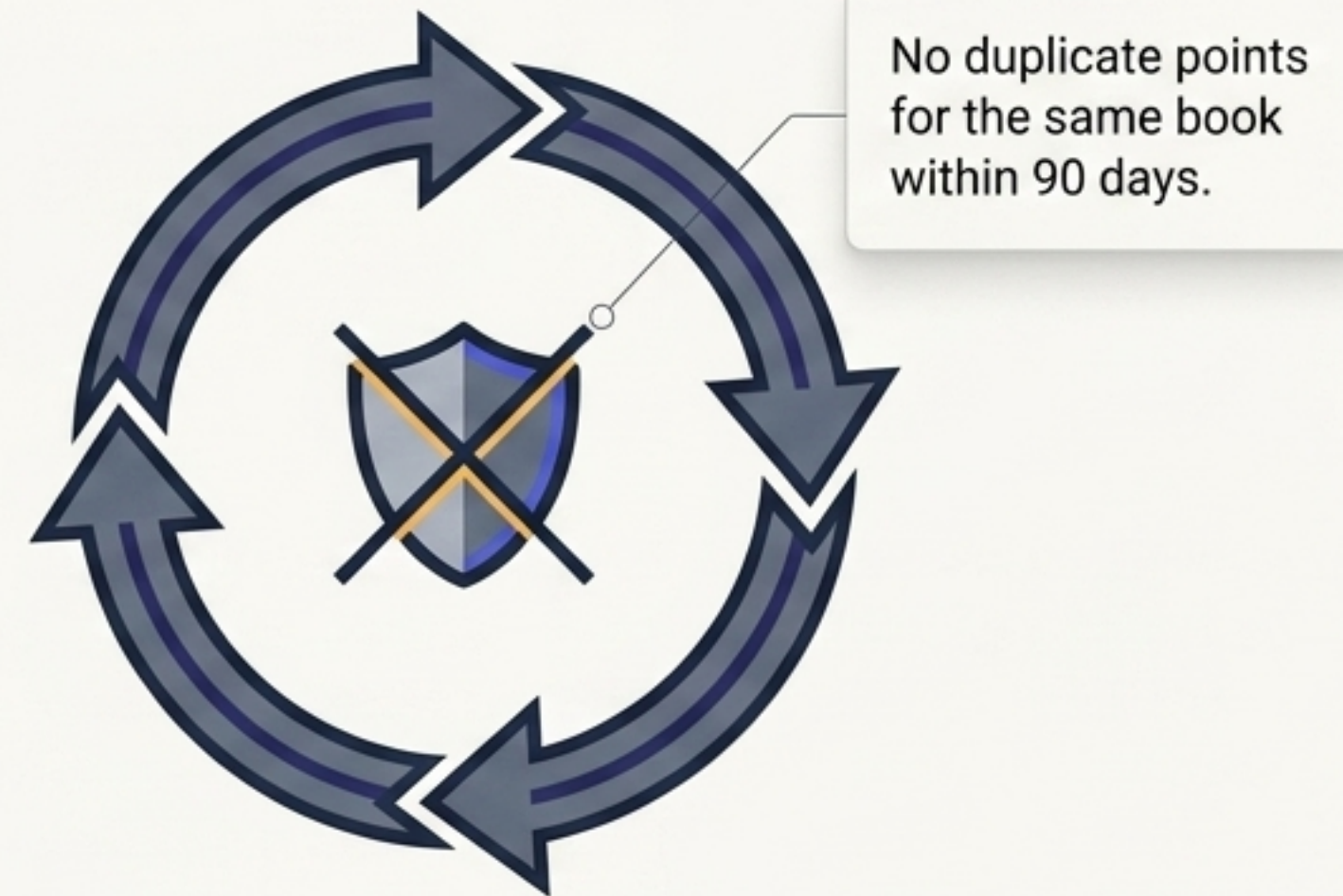
Anti-Exploit Security Locks

Algorithmic friction to ensure genuine reading and prevent point farming.

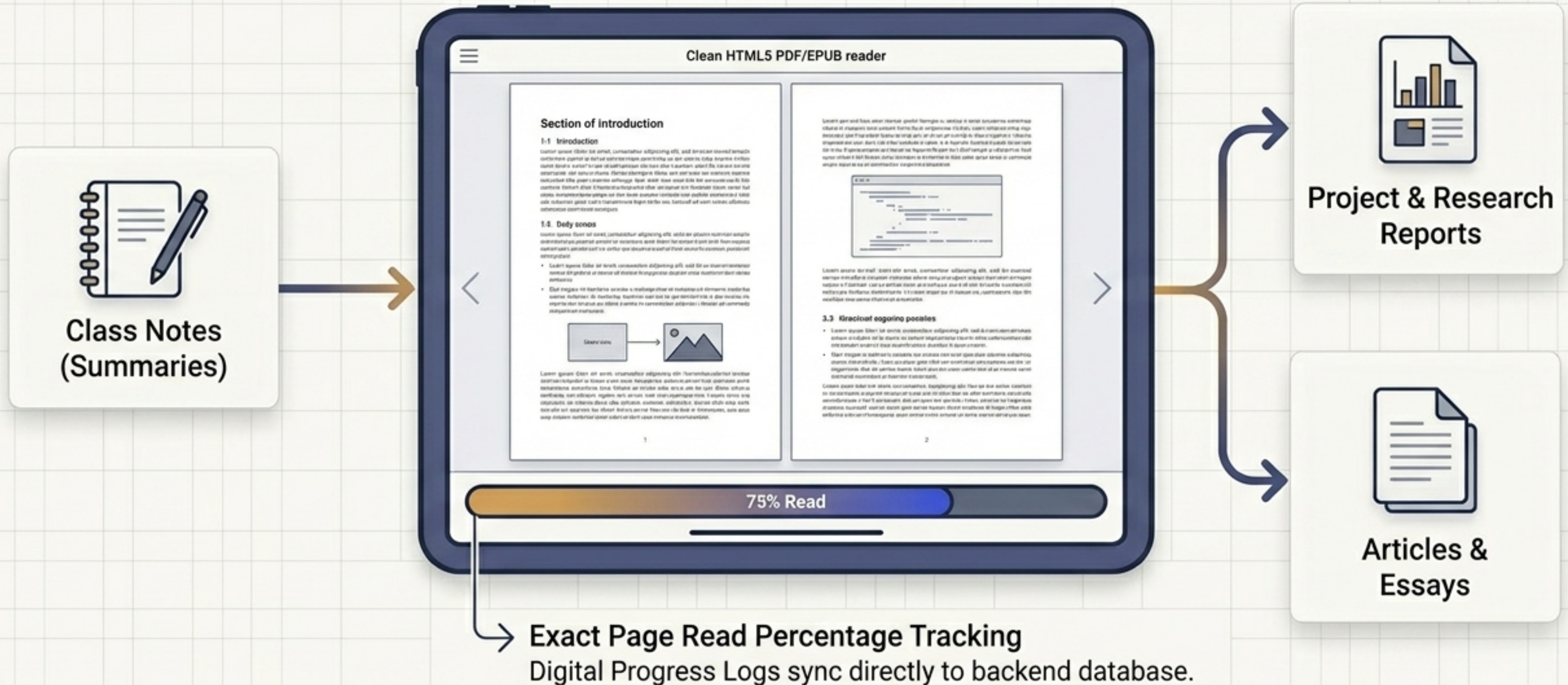
Minimum Reading Period Lock



The 90-Day Cooldown



Community e-Library & Peer Repository

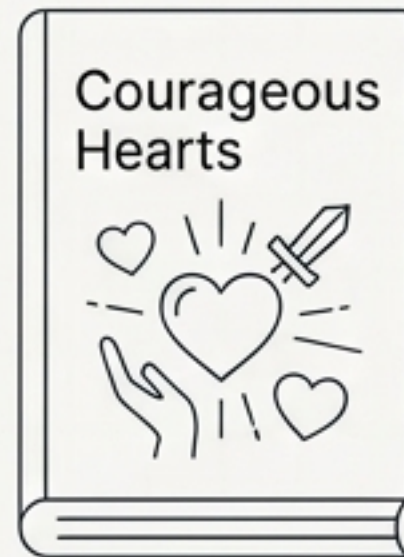


Intent-Based AI Search Engine

Local Hybrid Search

Instant JavaScript Filtering

🔍 courage in space



AI Semantic Search

NVIDIA Completions Intent Ranking

🔍 courage in space



The Librika Ecosystem Flywheel

How infrastructure translates to continuous educational engagement.

